

OzLand: Exploring Dynamic Generative Game Design with Multi-level Runtime Authorization

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Pre-play Generation



Gameplay Screen



Runtime On-demand Generation

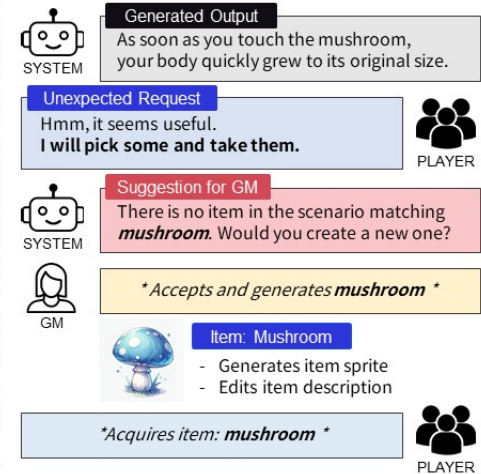


Figure 1: Overview of OzLand workflow. i) At the pre-play session, the human game master constructs the scenario and game components with a generative AI assistant, including interactive scenes and NPCs, objects, and game events involved in each scene. ii) The gameplay consists of text-driven role-play interaction. The player declares their action or talk with the NPCs by entering text input, while the LLM-driven system agent progresses the game and generates the output. iii) If an unprepared component is required during runtime, the system agent suggests that the GM pause the game and generate the new component.

Abstract

The advent of Generative AI has introduced new opportunities into game design by enabling the dynamic creation of content during gameplay. However, runtime generation of the game components faces two critical challenges: ensuring the quality of dynamically generated content and managing latency in real-time interactions. We present OzLand, a dynamic generative game co-directed by a human Game Master (GM) and a generative AI assistant. OzLand is a text-based role-playing game in which players freely interact

with the game through text inputs, and the LLM-based system progresses the game in response, interpreting players' actions and generating output dialogues. OzLand introduces two core strategies to ensure player experience while providing dynamically emerging content. First, we designate distinct roles for GMs and players, involving GMs to supervise content quality and trigger the generation of complex assets on demand at runtime. Second, we adopt a multi-level generation framework that separates pre-play and runtime sessions, enhancing lightweight generation during runtime and preparing resource-intensive content beforehand. OzLand explores a design approach empowering users to co-create emergent gameplay while mitigating the uncertainty of dynamic content generation.

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1 Introduction

The advent of generative AI has significantly shifted the paradigm of game development. Generative AI has been widely adopted to efficiently create multi-modal game assets, ranging from text and images to 3D objects, sound, and code. More significantly, generative AI enables game content to be dynamically created during runtime, fundamentally reshaping the gameplay experience. A growing number of works have explored the abilities of generating narratives [4, 8], agent behaviors [7], design components [1, 9], and even interactive scenes [2, 3] at runtime.

However, dynamic content generation is still constrained by two key challenges. First, it is difficult to guarantee the quality of the runtime-generated content. Generating satisfying multi-modal assets that align with the game’s context and mechanics is highly challenging and often requires iterative refinement under human authorization. Secondly, the delays of the generation degrade the gameplay experience, particularly lowering its feasibility for providing multi-modal assets in real-time.

We present OzLand, a dynamic generative game co-directed by a human Game Master (GM) and a generative AI assistant. OzLand is a text-based role-playing adventure game in which a generative agent functions as the underlying game system that interprets player inputs, progresses the scenario, and generates text and image assets. The players progress through the game by freely declaring actions or communicating with NPCs by text inputs. The LLM-driven game system generates text outputs and advances the game events according to the intended scenario.

2 System Design

Scene Image	Available events	Instruction
	Tea party with the mad hatter and the rabbit	The players visit a tranquil garden, where vibrant flowers are in full bloom. (...) At the table in the heart of the garden, an afternoon tea party is held.
	NPCs Mad hatter, rabbit	The players should be induced to join the tea party, talk with the NPCs, and receive hints about the location of the key to escape the prison.
	Interactive objects A cup of tea, table	
	Desired flow Participate in the tea party	

Figure 2: An example scene component construction during the pre-play session. A scene includes a background image sprite and a base description, along with lists of the IDs of game events, NPCs, and interactive objects that appear within the scene. In addition, the GMs are guided to provide instructions for the AI agent on how to direct players in order to progress the scenario as intended.

OzLand addresses the challenges of dynamic generation through mainly two strategies. First, OzLand distinguishes generation authorities between GMs and Players, encouraging GMs to supervise runtime-generated content and decide whether to run time-intensive generation. Inspired by TableTop Role-playing Game (TRPG), we involved a human GM as a unique role that constructs scenarios, supervises runtime-generated outputs, and co-directs the scenario through improvisation with the players. The players interact with the LLM-based system through their text inputs, but

are not granted the ability to generate scenario-related components. The GM participates in the game through a dedicated separate client, reviewing and confirming the generated output before it is presented to the players. In cases where the generated output deviates from the intended scenario, the GM may intervene and request regeneration.

Second, OzLand provides multiple levels of content generation to address generation latency and uncertainty. We divide content generation into pre-play and runtime sessions, encouraging lightweight generation (text and simple sprites) during runtime to maintain responsiveness. To achieve this, the LLM-driven assistant collaborates with the GM to predict and prepare time-intensive components required in the scenario, including interactive scenes, high-level narrative structures, and NPCs during the pre-play stage.

During gameplay sessions, the system primarily responds to player inputs with simple dialogues on each turn. The GM is provided 5 seconds to review the generated text before it is shown to the player and decide whether to confirm or reject it. For instance, if the generated output discloses narratives that are not yet supposed to be known to the player, the GM may reject the output. When unprepared components - such as new items or scenes - are required, the system proactively suggests to the GM for generation.

To minimize interruption time caused by runtime generation, the system produces four candidates for each generation. When the GM rejects the initial output, they are given the option to select an alternative from the remaining candidates. If no candidate is suitable, the GM can then request a regeneration with a revised prompt. Regenerating simple text outputs requires less than 10 seconds. Generating game components accompanying image generation takes 2-3 minutes on average (but potentially takes longer when multiple iterations are required).

3 Implementation

OzLand is composed of a player and GM client, connected to a core system server that integrates multiple generative AI modules. The player client features a 2D scene with a text input interface and dialogue output (the gameplay screen of Figure 1). The GM client extends the player client’s view with an editor interface for constructing scenarios and content generation during runtime.

The core system includes three main LLM modules utilizing GPT-4o [6] – the input interpreter, output generator, and GM assistant – along with a Dall-E 3 [5] image generation module. The **input interpreter** analyzes the player’s input to identify intended action and categorizes it under predefined system-level actions (e.g., moving to another scene). The **output generator** produces responses to player based on the predefined scenario and the current game progression. These responses include dialogue output shown to the player (i.e., narrative text and NPC dialogues) and system output interpreted automatically by the system (e.g., the ID of the event that will occur at the next turn). Finally, the **GM assistant** implements a conversational agent that supports the GM in both scenario construction during the pre-play stage and coordinating authorization during the runtime stage.

4 Future Work and Conclusion

We present OzLand, a dynamic generative game system co-directed by a human Game Master and an AI assistant. By developing OzLand, we investigate design strategies for managing the uncertainty of real-time content generation while supporting emergent gameplay properly authorized to align with the intended scenario. In future work, we aim to explore systems to support emergent runtime interaction beyond predefined game components, including player actions and unstructured in-game behaviors. We will also investigate interface design to assist non-expert GM users in efficiently managing complex dynamic game components.

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